

HTML

The Complete Book
Volume III: Advanced Tags

Every tag • Code + Output • Full colour

Er Sayan Mondal

Volume III • Advanced

Copyright

HTML — The Complete Book
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Preface to Volume III

Welcome to the **advanced** level. Here HTML starts doing serious work: images that adapt to screens, drawings made from code, clickable image regions, scripts, reusable templates and embedded media.

✅ **You should know:** Volumes I and II, plus how to run a page with Live Server.

📌 **Note:** This volume uses **only HTML**. A few tags (canvas, script, template) need JavaScript to do their full job. Here you learn what they are and how to write them; the JS book brings them to life.

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1 Responsive Images

Phones and big screens need different images. A small image saves data on phones; a large one looks sharp on desktops.

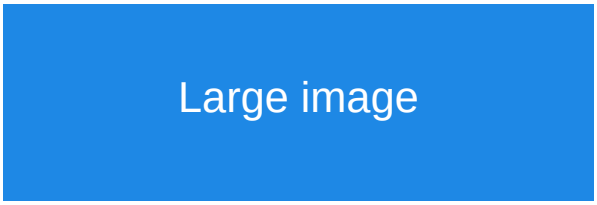
<picture> and <source>

`picture` wraps several `source` tags. The browser picks the **first** source whose `media` rule matches. The `img` inside is required (fallback + alt text).

CODE

```
<picture>
  <source media="(min-width:600px)" srcset="large.jpg">
  <source media="(max-width:599px)" srcset="small.jpg">
  <img src="fallback.jpg" alt="Beach view">
</picture>
```

OUTPUT



Wide screen shows blue, phone shows orange.

srcset and sizes on <img>

Give the browser a menu of files; it picks the best one for the screen.

CODE

```
<img src="photo-800.jpg"
     srcset="photo-400.jpg 400w, photo-800.jpg 800w, photo-1600.jpg 1600w"
     sizes="(max-width:600px) 100vw, 50vw"
     alt="Mountain">
```

Part	Meaning
400w	This file is 400 pixels wide
sizes	How wide the image will be on screen (100vw = full width)
type="image/webp"	On <source>: use modern formats with a fallback

Also add `loading="lazy"` to images below the fold, and always write width and height to avoid page jumping.

2 Inline SVG

SVG = Scalable Vector Graphics. Drawings written as tags. They stay sharp at any size and can be styled with CSS. Put them inside `<svg>`.

Tag	Draws	Key attributes
<rect>	Rectangle	x y width height rx
<circle>	Circle	cx cy r
<ellipse>	Oval	cx cy rx ry
<line>	Line	x1 y1 x2 y2 stroke
<polygon>	Closed shape	points
<path>	Any curve	d
<text>	Text	x y font-size

Basic shapes

CODE

```
<svg width="300" height="120" viewBox="0 0 300 120">
  <rect x="10" y="10" width="80" height="60" rx="10" fill="#4db8ff"/>
  <circle cx="150" cy="40" r="30" fill="#f7a34b"/>
  <polygon points="220,70 250,10 280,70" fill="#1e88e5"/>
  <line x1="10" y1="100" x2="290" y2="100" stroke="#0b3a6e" stroke-width="4"/>
  <text x="10" y="118" font-size="14" fill="#0b3a6e">SVG shapes</text>
</svg>
```

OUTPUT



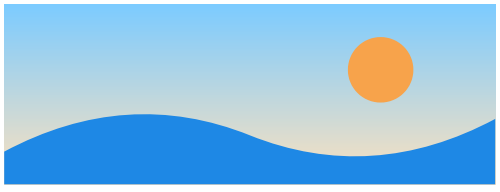
SVG shapes

Path and gradient

CODE

```
<svg width="300" height="110" viewBox="0 0 300 110">
  <defs><linearGradient id="sky" x1="0" y1="0" x2="0" y2="1">
    <stop offset="0" stop-color="#7ccbff"/><stop offset="1" stop-color="#ffe2bd"/>
  </linearGradient></defs>
  <rect width="300" height="110" fill="url(#sky)"/>
  <circle cx="230" cy="40" r="20" fill="#f7a34b"/>
  <path d="M0 90 Q75 50 150 80 T300 70 V110 H0 Z" fill="#1e88e5"/>
</svg>
```

OUTPUT



viewBox="0 0 300 110" defines the drawing's own coordinate system. It is what makes SVG scale.

3 Canvas (the tag)

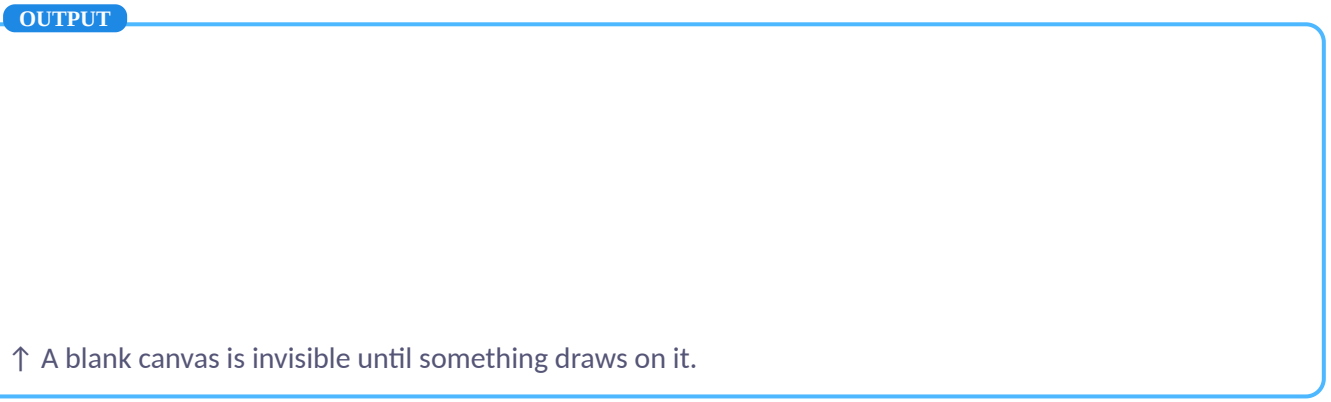
`<canvas>` creates a blank rectangular drawing board inside your page.

The tag only **makes the board**. Painting shapes, text and animation on it needs JavaScript, which you will learn in the JS book. Here you learn to write the tag correctly.

CODE

```
<canvas id="board" width="300" height="140">Your browser does not support canvas.
</canvas>
```

OUTPUT



↑ A blank canvas is invisible until something draws on it.

Part	Meaning
width / height	Size of the board in pixels
id	A unique name so it can be found later
Text inside	Fallback shown by very old browsers

SVG (Chapter 2) is drawn by writing tags, so you can already make pictures. Canvas is painted by code.

4 Image Maps

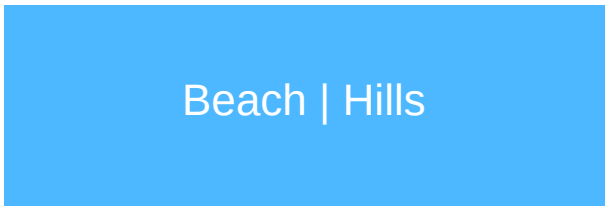
An **image map** makes different parts of one image clickable. Three pieces work together: an `img` with `usemap`, a `<map name>`, and `<area>` tags that mark the hot spots.

area shape	coords
rect	x1,y1,x2,y2 (top-left and bottom-right)
circle	x,y,radius
poly	x1,y1,x2,y2,x3,y3…

CODE

```
<img src="places.png" usemap="#places" alt="Beach and hills" width="300"
height="100">
<map name="places">
  <area shape="rect" coords="0,0,150,100" href="https://example.com/beach"
target="_blank" alt="Beach">
  <area shape="rect" coords="150,0,300,100" href="https://example.com/hills"
target="_blank" alt="Hills">
</map>
```

OUTPUT



Click the left half or the right half. Coordinates are in image pixels.

Image maps do not resize with the image. For clickable graphics that scale, use SVG.

5 noscript, template, script, link, base

<noscript>

Content shown only when JavaScript is turned off.

CODE
```
<noscript>Please turn on JavaScript to use this site.</noscript>
```

OUTPUT

(Invisible here because JavaScript is on in your browser.)

<template>

Holds HTML that is **never shown** by itself. JavaScript can copy it onto the page later.

CODE
```
<template>
  <li>A ready-made list item</li>
</template>
```

<script>

The tag that attaches JavaScript to a page. You only need to recognise it now: `src` points to a JS file. Writing that file is the job of the JS book.

CODE
```
<script src="app.js"></script>
```

<link> and <base> (in head)

Tag	Example	Use
<link>	rel="icon" href="favicon.png"	Tab icon
<link>	rel="canonical" href="https://site.com/page"	Main address of the page
<link>	rel="preload" href="hero.webp" as="image"	Load an important file early
<base>	href="https://site.com/" target="_blank"	Default base for all links

6 Media, embed, object, iframe

<video> with subtitles

Multiple `source` formats + a `track` for subtitles (a .vtt file). Attributes: **controls poster preload="none|metadata|auto" autoplay muted loop playsinline**.

CODE

```
<video controls width="320" poster="cover.jpg" preload="metadata">
  <source src="trip.webm" type="video/webm">
  <source src="trip.mp4"  type="video/mp4">
  <track src="en.vtt" kind="subtitles" srclang="en" label="English" default>
  Your browser does not support video.
</video>
```

Browsers block autoplay with sound. Use `autoplay muted` together.

<iframe>

Embeds another page. **sandbox** restricts it for safety, **loading="lazy"** delays loading, **allow** grants features (fullscreen, camera), **srcdoc** holds inline HTML.

CODE

```
<iframe srcdoc="<h3 >Hello from inside an iframe</h3>"
        width="100%" height="70" sandbox title="Demo frame"></iframe>
```

OUTPUT

Hello from inside an iframe

<embed> and <object>

Older ways to include PDFs, SVG or plugins. Prefer `iframe` or `img` today.

CODE

```
<object data="brochure.pdf" type="application/pdf" width="300" height="200">
  <a href="brochure.pdf">Download the PDF</a>
</object>
<embed src="drawing.svg" type="image/svg+xml" width="100" height="100">
```

7 Mini Project: Travel Gallery Page

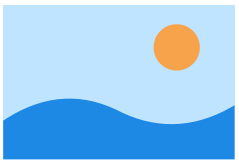
Build a small page using Volume III tags only: **figure, svg, picture, map and iframe** together with earlier tags.

CODE

```
<article>
  <h3>Goa Getaway</h3>
  <figure>
    <svg width="120" height="80" viewBox="0 0 120 80">
      <rect width="120" height="80" fill="#bfe4ff"/>
      <circle cx="90" cy="22" r="12" fill="#f7a34b"/>
      <path d="M0 60 Q30 40 60 55 T120 52 V80 H0 Z" fill="#1e88e5"/>
    </svg>
    <figcaption>Sunset at the beach</figcaption>
  </figure>
  <p>Departure: <time datetime="2026-12-25">25 Dec 2026</time></p>
  <details>
    <summary>What is included?</summary>
    <ul><li>Hotel and breakfast</li><li>Beach tour</li></ul>
  </details>
  <a href="https://example.com/book" target="_blank" rel="noopener">Book now</a>
</article>
```

OUTPUT

Goa Getaway



Sunset at the beach

Departure: 25 Dec 2026

▶ What is included?

Book now

✍️ Practice

1. Add a second card for Manali with your own SVG picture.
2. Add a `picture` tag with two different images.
3. Embed a page using `iframe` with `srcdoc`.

Advanced level done! Volume IV covers pro topics: SEO, accessibility (ARIA), performance and security. 🚀

Keep Building! 

"Every expert developer once typed <h1>Hello World</h1>."

Er Sayan Mondal
Author

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